

CULTURED

Complete Product & Technical Architecture

Version 4.0 — from local-storage prototype to market-ready, zero-budget, scalable dating app. Matching on humor + music first, faces second.

- \$0 INFRASTRUCTURE
- SCALABLE BY DESIGN
- BUILDS ON V3 PROTOTYPE
- STORE-READY CHECKLIST

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Inputs analyzed: cultured v3.html front-end prototype · v1 blueprint PDF · founder's launch notes

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1 · Product Vision, Positioning & What Changes in v4

Vision (unchanged, restated crisply): Cultured matches people on how they think and what they love — humor style and musical taste — before it weighs a face. Every match starts from a shared laugh or a shared track, never a profile photo alone. Target: Gen Z / young millennials burned out on photo-first swiping who want proof of compatibility before the first message.

Positioning (validated by the v1 competitive analysis, still true):

- **vs. Schmooze** — memes *and* music together, so humor-only false positives (funny but sonically incompatible) are filtered out. Content browsing and partner evaluation are separate surfaces from day one.
- **vs. Turn Up / Vinyly** — a genuinely free core loop: no paywall on seeing likes or sending first messages, and an active daily ritual instead of passive scrobbling.
- **vs. Boo** — signals people generate every day (meme reactions, saves, co-listens), not a one-time personality quiz that goes stale; mandatory verification from day one.

1.1 What the v3 prototype already proves

The v3 HTML prototype is a complete, polished front-end running on fake data and localStorage. It already implements: onboarding with taste calibration ("Pick a few starting tastes", "What brings you here?"), the Culture Feed with "Today's memes", the Daily Drop ritual ("Drops at 9:00 AM"), the Match Matrix deck with resonance moments ("You and {name} resonate"), the Cultural Fingerprint profile (meme shelf, artists, playlists, prompts, anti-genres, humor signals), Circles ("From your circles"), chat threads, 15-minute listening sessions ("That's your 15 minutes"), settings with privacy and data controls. **v4 keeps this UX intact — the work is everything behind it.**

1.2 The six upgrades that define v4

#	UPGRADE	WHY IT'S THE RIGHT NEXT MOVE
1	Real backend & identity — phone + Apple/Google sign-in, one database for profiles, matches, messages; live chat replaces localStorage.	Blocker #1 from the founder's list. Nothing is shared between users today; without this nothing else matters.
2	The Resonance Engine — the matching algorithm becomes a real, versioned service: humor vector + music vector → one Taste Twins score per candidate pair.	The prototype fakes compatibility. This is the product's entire promise, so it gets its own rigorously-specified pipeline (§6).
3	Spotify/Apple Music OAuth — taste pulled from real listening history at signup; manual chips remain as fallback.	Resonance scores need real data behind them; OAuth import kills the cold-start problem in one step.
4	Real meme pipeline — licensed/UGC daily feed + "post your own meme" flow with moderation; sending memes to matches becomes the core loop.	Memes are the icebreaker currency; they must be legally safe (DMCA) and fresh daily.
5	Safety stack — mandatory selfie-liveness verification, enforced 18+ gate, report/block, content moderation pipeline, privacy policy / ToS / account deletion.	App stores reject dating apps without these; Boo's scammer problem shows the trust cost of skipping them.
6	Native shell + push — wrap the existing web app with Capacitor for App Store / Play listings, push notifications (Daily Drop at 9:00), haptics.	The v3 UI is already mobile-web; Capacitor turns it into a store binary in days, not a rewrite.

1.3 New ideas brainstormed for v4 (on-vision only)

Every idea below serves the existing thesis — humor + music compatibility — and extends the v1 blueprint's brainstorm list. Nothing generic (no video dating, no games-for-games'-sake).

Deepen the cultural signal

- **Meme Duel** NEW — two matched people independently caption the same 5-meme set; a "humor compatibility" card reveals both caption sets side by side. Built-in comedic tension, extremely screenshot-able for Stories.
- **Anti-genre vector** NEW — dislikes are first-class: flagging "never show me country / cringe humor" subtracts from the score, because mismatched taste predicts incompatibility more than shared taste predicts compatibility.
- **Taste Drift timeline** — private, opt-in view of how your humor/music vectors moved over months. Nostalgia hook, never shown to matches.
- **Concert radar** — surfaces matches who follow the same artist with an upcoming show in your city (free Bandsintown/Ticketmaster Discovery API). Turns a match into a real-world plan.

Growth & shareability

- **Taste Card** NEW — a gorgeous server-rendered image of your Fingerprint (top artists, humor radar, best memes) with a watermark + "find who else relates" CTA. Best free growth channel; every share is an ad.
- **Daily Drop + streak** — the prototype's 9:00 AM drop becomes a streak ritual with a fixed-time push. Reacting feeds the vector model even on days you don't browse.
- **Watermarked external shares** — memes shared to Instagram/WhatsApp carry a subtle logo + CTA (only externally, never in-app).
- **Squad remix** — remix a friend's meme with your own caption inside Circles; spreads the app through friend groups, not just daters.

Kill first-message anxiety

- **Icebreaker Roulette** — if a new match sits unopened 30+ seconds, suggest an opener built from the single most-overlapping niche meme/artist.
- **Vibe-check voice notes** — 10-second audio reaction to a shared track; lower effort than typing, higher signal than a like.
- **Co-listen countdown** — "Listen together at 8pm?" quick-reply chip schedules a session with an in-chat reminder.
- **Meme-reply threads** — reply to a specific meme in chat with emoji reactions; read receipts (premium).

Trust & quality

- **Verified Fingerprint badge** — separate from photo verification: confirms music/meme data comes from a real, aged OAuth account, not bot engagement.
- **Vibe-report** — a lightweight "this didn't feel like a match" signal distinct from abuse reporting; quietly recalibrates the algorithm without punishing the reported user.
- **Date-share safety** — one tap shares a match's profile + meetup plans with a trusted friend.

Deliberate non-features (carried from v1, still binding): no swipe-to-judge-a-person gesture anywhere in the Culture Feed; no "who liked you" paywall; no guilt-trip notifications ("3 people are waiting!"); no grid of faces — the Match Matrix stays a calm, single-card deck.

2 · Full Feature Set — v4 Specification

2.1 Onboarding & verification (first-run funnel)

STEP	SCREEN	PRODUCTION BEHAVIOR
1		3 slides pitching the Fingerprint concept before any ask: "Match on your humor, not your headshot."

	Welcome carousel	
2	Auth + age gate	Phone OTP or Sign in with Apple / Google (both required by App Store rules). Mandatory 18+ date-of-birth check, enforced server-side — an under-18 signup is hard-blocked, not warned.
3	Photo verification	Real-time selfie liveness (on-device ML Kit face + liveness heuristics) matched against profile photos later. Mandatory before the profile is visible in the Match Matrix.
4	Music connect	Spotify OAuth (primary) / Apple Music (phase 2). One pull of top artists/tracks/genres seeds the music vector. Skippable → manual genre-tag fallback.
5	Meme calibration	15–20 rapid meme swipes (laugh / meh / skip) seeding the humor vector. This is the prototype's "Pick a few starting tastes" made continuous.
6	Music calibration	10 × 7-second clips across genres, swipe to react. Cross-references the OAuth import; required if step 4 was skipped.
7	Photos + prompts	2–6 photos (equal-weight to Fingerprint, not the hero), 3 of 8 cultural prompts ("a movie I'll defend to the death", "my most-played song at 16").
8	Basics	Name, age, location, gender/orientation, looking-for (Dating / also-open-to-Friends dual mode).
9	Permissions	Notifications + location asked LAST, after value is shown. Daily Drop push opt-in framed as "get your 9 AM meme".

Design intent: verification and calibration come before bio fields, so the algorithm has real signal the moment a user reaches the Match Matrix — no cold-start blank profile.

2.2 Cultural Fingerprint (profile)

- **Fingerprint tab (default landing):** humor-style radar (affiliative / self-enhancing / aggressive / self-defeating — the four Humor Styles Questionnaire axes) auto-computed with a plain-language line ("leans affiliative and self-enhancing"); top 5 artists + top 3 genres pulled live, auto-refreshing; 3 pinned playlists (30-sec inline previews); "Recently vibed with" — last 5 reacted memes as a curated shelf; grayed-out Taste Twins preview ("unlocks when you match").
- **Photos tab:** image carousel, verification badge as a small check, not a banner.
- **Prompts tab:** 3 text/audio prompts, deliberately cultural.
- **Edit mode:** manual unpin/repin of playlists and memes overrides auto-curation.
- **Privacy toggle:** "show my Fingerprint to matches only" vs. visible in Discovery previews.

2.3 Culture Feed (daily habit surface — never any person-evaluation UI)

- **Meme cards** — like, laugh-react (distinct signal mapped to the humor model), comment, share-with-caption to a match or Circle.
- **Track cards** — 15-sec preview, save-to-Fingerprint, "who else vibes with this" tap-through.
- **Daily Drop card** — pinned top once/day at 9:00 AM local, one meme + one track, higher weight in the vector update; streak counter ("🔥 12 days").
- **Local Discovery card (opt-in)** — "3 people near you liked this exact meme", soft CTA into the Match Matrix, never an inline swipe.
- **Meme Duel card (new)** — weekly duel prompt; entering pairs you with a match for the caption duel.

2.4 Match Matrix (calm, single-card deck)

- **Card anatomy:** photo, name, age, distance + **Taste Twins badge** ("87% cultural match") + one shared-artist chip + one shared-meme-category chip as the built-in icebreaker hook. Expand to full Fingerprint before deciding — no forced blind swiping.
- **Actions:** like, pass, **Resonate** (super-like, 1/day free), rewind (premium).
- **Enhanced Profile Discovery:** tap a shared niche meme/artist → "12 local singles who also loved this" — any piece of content becomes its own matchmaking entry point.
- **Dual mode:** persistent Dating / Friends toggle sharing one Fingerprint and algorithm, different candidate pool and card copy.
- **Empty state:** never a dead end — "come back to the Culture Feed, your Fingerprint gets stronger the more you're there".

2.5 Chat & Listening Sessions

- **Chat:** text, image, voice notes; one shared meme/track auto-attached from the match moment so the first message never starts from zero; meme-reply threads; emoji reactions; read receipts (premium).
- **Listening Session:** either user picks a track/short playlist → WebSocket-synced play/pause/seek across both devices → shared waveform scrubber with timestamped emoji bursts → auto-logs to the match's private "**Our Soundtrack**". Free tier caps at 15 minutes (as prototyped: "That's your 15 minutes"); Playlist Pass removes the cap. WebRTC reserved for opt-in live voice.
- **Music trivia:** 5-question blind-test built from tracks both Fingerprints share; 2 free plays/day, unlimited on premium.
- **Meme Duel (new):** both caption the same 5 memes independently → reveal card with humor-compatibility verdict → shareable image.

2.6 Circles (squad layer)

Friend groups that share and remix memes ("From your circles" in v3). Remix-with-caption re-shares spread the app inside existing friend groups — widening the funnel to people browsing before they're ready to date. Circle activity notifications are kept strictly separate from match activity.

2.7 Notifications & Settings

- **Three-section activity tab:** Matches & messages · Culture Feed activity · Fingerprint updates (light, dismissible nudges only, e.g. "Your humor profile just got sharper").
- **Push categories, separately toggleable:** matches & messages (default ON), Feed activity, Daily Drop, marketing (default OFF).
- **Settings:** discovery preferences; reconnect/disconnect Spotify; wipe-and-rebuild humor vector (real reset); export my data; anti-genre list; Fingerprint visibility; incognito (premium); block/report management; re-run verification; subscription & IAP management; Safety Center; deactivate (soft) vs. delete (hard, with cooling-off window — required by both stores).

3 · Technology Stack — \$0/Month, Chosen for Scale

Constraint: no investment — every pick below has a genuine free tier that covers launch, and a clean scaling path when a tier is outgrown. Nothing chosen here needs to be re-architected at 10k–100k users; only plans change, not code.

3.1 The stack at a glance

LAYER	CHOICE	FREE TIER COVERS	WHY THIS ONE
Client	Existing v3 web app, modularized → wrapped with Capacitor (iOS + Android)	100% free (open source)	The prototype IS the UI — Capacitor turns it into store binaries with push, haptics, native OAuth. No React Native rewrite at launch; RN remains an option later (the v1 plan) once revenue exists.
Backend-as-a-Service	Supabase — Postgres, Auth (phone OTP, Apple, Google), Realtime (websockets), Storage, Edge Functions (Deno)	500 MB DB, 1 GB storage, 2 GB bandwidth, 50k MAU auth, 200 concurrent realtime connections, 500k edge-function invocations	One platform replaces a hand-built API server, socket server, auth service and file store. Postgres underneath = real SQL, real constraints, pgvector for the matching engine — no separate vector DB bill.
Matching engine	pgvector inside Supabase Postgres + Edge Functions for vector updates	Included in DB	Humor & music vectors are ~64–128-dim floats; cosine similarity in SQL with an IVFFlat/HNSW index handles 100k+ users at millisecond speed. No Python ML service to host at launch.
Meme understanding	Phase 1: human-tagged taxonomy (30 humor categories × 4 style axes) applied at upload/moderation. Phase 2: CLIP + OCR embeddings via Hugging Face Inference (free tier) or a \$5 VPS cron	HF free tier ~rate-limited but sufficient for daily batch	A curated taxonomy beats a half-trained model at cold-start, costs nothing, and the tag data it produces becomes the training set for the CLIP upgrade (v1 architecture's ViT/CLIP/Sentence-BERT vision — same destination, cheaper road).
Music data	Spotify Web API (OAuth, top artists/tracks — free) + Last.fm API (genre/tag enrichment, recommendations)	Both free; Last.fm ~2 req/s → handled with backend cache + async queue	Exactly the v1 hybrid: Spotify for the authoritative one-time pull, Last.fm as the ongoing enrichment engine (Spotify Extended Quota needs 250k MAU — irrelevant at launch).
Push notifications	Firebase Cloud Messaging via Capacitor (APNs bridged)	Unlimited, free	Industry standard; the 9:00 AM Daily Drop is a scheduled Edge Function → FCM fan-out.
Media storage & images	Supabase Storage (origin) + Cloudflare free CDN in front; image resize via Edge Function or Cloudflare free rules	1 GB storage, Cloudflare free CDN unlimited bandwidth	Memes/photos served through Cloudflare = the 2 GB Supabase bandwidth cap effectively never bites.
Taste Card / OG images	Edge Function rendering SVG → PNG (Deno + resvg) — no paid screenshot service	Within edge invocations	Share cards must be generated per-user on demand; server-side SVG keeps them crisp and free.
Moderation		Launch-scale volumes	

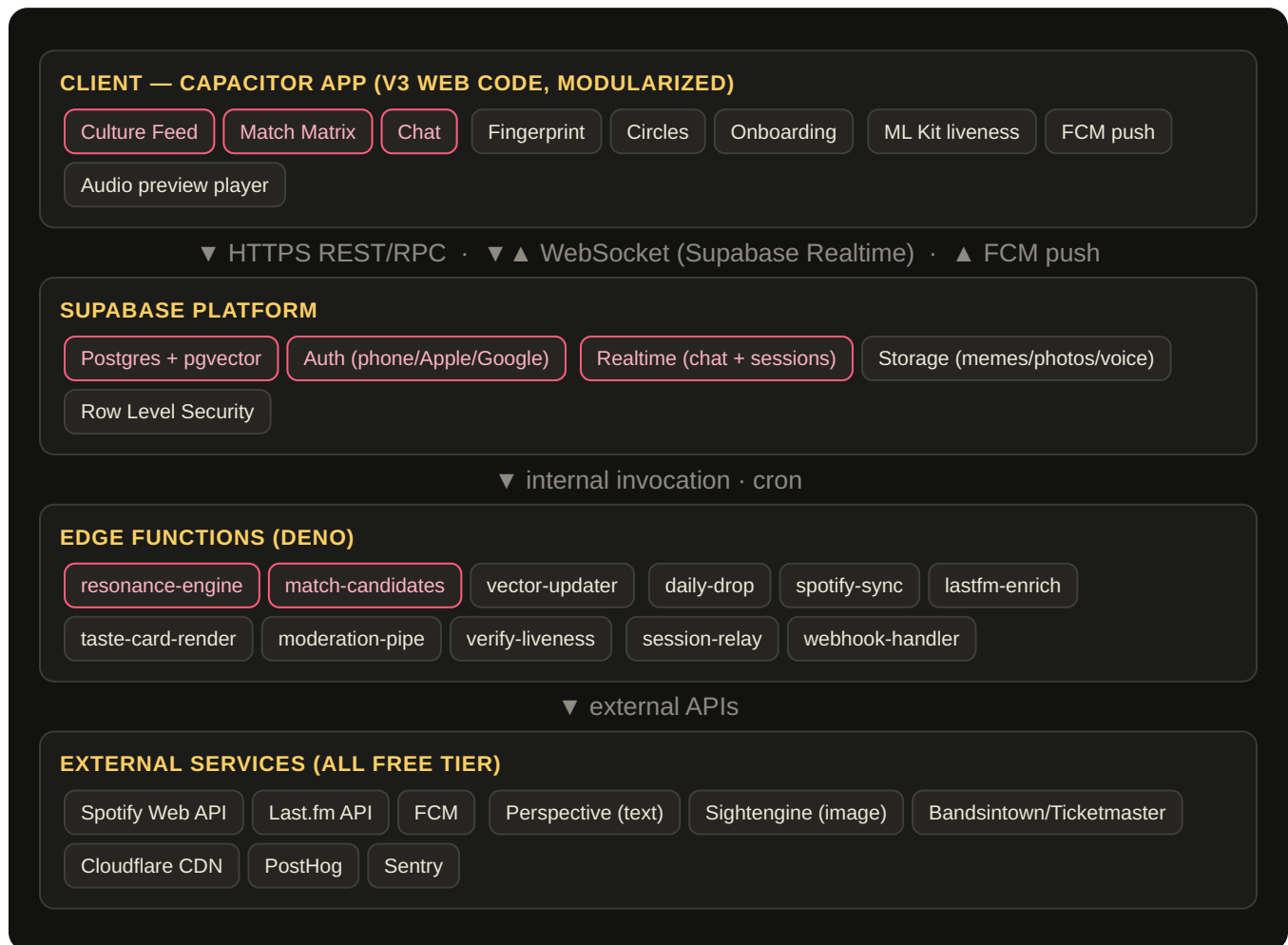
	Google Perspective API (text toxicity, free) + Sightengine (image nudity/violence, free ~2k ops/mo) + human review queue		Automated first pass, human second pass for reports; both stores demand a working moderation story.
Liveness / verification	Google ML Kit on-device (free, offline) face detection + liveness challenge (blink/turn), selfie ↔ profile similarity via on-device FaceNet embedding compared in an Edge Function	100% free (on-device)	Paid vendors (Veriff etc.) are the phase-2 upgrade; ML Kit gets a real, enforced check at \$0.
Analytics	PostHog open-source cloud free tier (1M events/mo) or self-hosted	Launch scale	Funnel + retention analytics without Firebase lock-in; event schema defined in \$7.4.
Error/crash	Sentry free tier (5k errors/mo)	Launch scale	—
Code & CI	GitHub + GitHub Actions	Free for the repo size	Type-check, lint, build, deploy Edge Functions on push.
Apple Music (phase 2)	MusicKit JS — free with Apple Developer account	—	Deferred: Spotify covers the majority of the target demographic at launch.

3.2 Why not the alternatives

- **Firestore:** equally valid BaaS, but Firestore can't do vector similarity or relational match queries without a paid vector add-on; Supabase's Postgres is the better match for a matching engine. (FCM is still used for push — it's free and orthogonal.)
- **Custom Node/socket server on a VPS:** \$0 only until you count your time; you also rebuild auth, storage and realtime that Supabase gives free. Revisit at scale.
- **React Native rewrite now:** discards a polished, working UI. Capacitor ships the same code; RN is a post-revenue performance/nativeness decision, not a launch blocker.
- **Pinecone/Weaviate:** a separate vector DB is a bill and an ops burden; pgvector is inside the database you already have.

4 · System Architecture

4.1 Component map



4.2 The one feedback loop that matters

The v1 blueprint's core insight stands: the three pillars are **one continuously-updating system**, not three features. In v4 it becomes an explicit event pipeline:

```
Feed interaction (laugh / save / skip / duel / co-listen)
  |
  ▼
events table (append-only, the source of truth)
  |
  ▼
vector-updater Edge Function (per-event, <100 ms)
  |   humor_vec ← decay(humor_vec) + w·meme_style_vec
  |   music_vec ← decay(music_vec) + w·artist_genre_vec
  ▼
fingerprints row updated (pgvector columns, incremental – never batch)
  |
  ▼
match-candidates re-ranks lazily on next Match Matrix open
  |   candidate score = TasteTwins(me, candidate) (see §6)
  ▼
Match Matrix deck – Taste Twins badge + shared-artist/meme chips
```

Sizing rule from v1, honored: humor and music vectors update on every Feed interaction, so the pipeline supports frequent, cheap incremental updates rather than periodic batch recomputation. One SQL UPDATE per event; no nightly job on the critical path.

4.3 Realtime design

FEATURE	TRANSPORT	DESIGN
Chat messages	Supabase Realtime (postgres_changes on messages) + broadcast for typing indicators	RLS limits subscriptions to thread participants; optimistic UI send with server confirm.
Listening Session sync	Realtime broadcast channel per session	Host-authoritative clock: host emits {t_ms, playing, track_id} every 2s + on seek; guests apply with 300ms drift correction. Emoji bursts are timestamped broadcast events persisted to session_moments .
Meme Duel	REST submit + realtime reveal	Captions stored server-side; reveal event fires only when both submitted — no peeking.
Presence	Realtime presence API	"online now" dots; off by default for privacy.
WebRTC voice (phase 2)	WebRTC data/audio, signaling over Realtime	Only when both opt in during a session.

4.4 Caching & rate-limit strategy

- **Last.fm (~2 req/s):** all enrichment through a queue table (enrich_jobs) drained by a 1-per-400ms cron Edge Function; results cached in artist_tags / track_tags keyed by MBID/artist+track so any artist is fetched once, ever.
- **Spotify:** top-artists pull once at signup + optional 30-day refresh; tokens stored encrypted (pgsodium), refresh handled server-side.
- **Media:** Cloudflare in front of Supabase Storage; meme images resized to 3 variants at upload.
- **Candidate ranking:** per-user top-200 candidate list materialized on Matrix open, cached 15 min in candidate_cache ; invalidated early on new match or filter change.

5 · Data Model — PostgreSQL Schema

Core tables. All tables have `id uuid pk default gen_random_uuid()`, `created_at timestamptz default now()`, and Row Level Security enabled. Vectors use `vector(64)` (humor) / `vector(128)` (music).

```

-- IDENTITY -----
users          id, phone citext unique, email, dob date,          -- 18+ enforced by CHECK + trigger
               gender, orientation text[], looking_for text[],    -- dating / friends
               city, lat, lng geog, verified_photo bool default false,
               verified_fingerprint bool default false,          -- OAuth-aged-account badge
               mode_default text, tier text default 'free',        -- free / premium
               deleted_at, deactivated_at
profiles       user_id fk-users, display_name, bio, photos jsonb, -- 2-6 urls, order preserved
               prompts jsonb,                                     -- [{prompt_id, answer_text|
audio_url[]]

               fingerprint_visibility text default 'matches_only'

-- TASTE -----
fingerprints   user_id fk pk, humor_vec vector(64), music_vec vector(128),
               humor_style jsonb,                                -- {affiliative:.7, self_enhancing:.6, aggressive:.1,
self_defeating:.2}
               anti_genres text[], top_artists jsonb, top_genres text[],
               pinned_playlists jsonb, pinned_memes uuid[], vector_version int, updated_at
events         id, user_id fk, kind text,                        -- laugh|like|skip|save|share|duel|listen|drop|react|
vibe_report    target_type text, target_id uuid, weight numeric, meta jsonb
               -- append-only source of truth for vector updates + analytics

-- CONTENT -----
memes          id, source text,                                  -- ugc | licensed | pack
               image_url, thumb_url, caption_text, ocr_text,
               style_vec vector(64),                             -- humor-style embedding of the meme
               categories text[], license_meta jsonb, status text default 'pending', -- pending|live|
removed        uploader_id fk nulls ok
tracks         id, spotify_id unique, title, artist, preview_url, genres text[], embedding vector(128)
artist_tags    artist text pk, tags jsonb, fetched_at          -- Last.fm cache
track_tags     track_key text pk, tags jsonb, fetched_at        -- Last.fm cache
daily_drops    day date pk, meme_id fk, track_id fk
reactions      user_id fk, meme_id fk, kind text,              -- laugh|like|meh
               unique(user_id, meme_id, kind)
comments       id, meme_id fk, user_id fk, body, status text
packs          id, title, assets jsonb, price_cents, licensed bool

-- MATCHING -----
likes          from_user fk, to_user fk, kind text,             -- like|resonate
               via text,                                         -- matrix|meme|artist|drop
               created_at, unique(from_user, to_user)
matches        id, user_a fk, user_b fk, taste_twins int,      -- score at match time (snapshot)
               shared_artist text, shared_meme_cat text,        -- icebreaker chips
               source text, unmatched_at
vibe_reports   from_user fk, to_user fk, reason text           -- recalibration signal, NOT abuse
blocks         blocker fk, blocked fk, created_at

-- CHAT & SESSIONS -----
threads        id, match_id fk unique, last_msg_at
messages       id, thread_id fk, sender_id fk, kind text,      -- text|image|voice|meme|track
               body, media_url, reply_to uuid, read_at
sessions       id, match_id fk, started_by fk, started_at, ended_at, seconds int
session_moments id, session_id fk, user_id fk, track_id, t_ms int, emoji
soundtracks    match_id fk, track_id fk, added_at,             -- "Our Soundtrack"
               primary key (match_id, track_id)
duels          id, match_id fk, meme_set uuid[], status        -- waiting|revealed
duel_captions  duel_id fk, user_id fk, captions text[],        -- hidden until both submit
               submitted_at, unique(duel_id, user_id)

-- SOCIAL -----
circles        id, name, created_by fk
circle_members circle_id fk, user_id fk
circle_posts   id, circle_id fk, user_id fk, meme_id fk, caption, remix_of uuid null

```

```

-- OPS -----
reports      id, reporter fk, target_type, target_id, reason, detail, status,  -- ABUSE path
              priority int, created_at
moderation_queue id, item_type, item_id, auto_score numeric, auto_flags text[],
              reviewer, decision, decided_at
notifications  id, user_id fk, category text, payload jsonb, read_at      -- category = per-type toggle
push_tokens    user_id fk, token text, platform text
subscriptions  user_id fk, product_id, store text, status, expires_at
entitlements    user_id fk, key text, value jsonb      -- playlist_pass, meme_packs, rewind credits

```

5.1 Security model (RLS, non-negotiable)

- Every table denies by default; policies grant `auth.uid() = user_id` (own rows) or match membership (`is_match(user_a, user_b)` security-definer function) for threads/messages/sessions.
- `fingerprints`: visible to matches; pre-match visibility only when `fingerprint_visibility='public'`, and only the display-safe projection (never raw vectors).
- `events`: insert-only from clients, read never — vectors can't be reverse-engineered from the API.
- `likes`: a user sees only likes they sent; incoming likes revealed only on mutual match (this is how "no paywall on seeing likes" stays true *and* scrape-proof).
- Storage buckets: `photos` (authenticated read), `memes` (public via CDN), `voice` (match-scoped signed URLs, 1h expiry).

6 · The Resonance Engine — Matching Algorithm Spec

6.1 Vectors

- **Humor vector (64-d)**: dims 0–3 = the four humor-style axes (affiliative, self-enhancing, aggressive, self-defeating); dims 4–33 = 30-category meme taxonomy affinity (absurd, deadpan, wholesome, dark, surreal, relatable, niche-hobby, wordplay, anti-humor...); dims 34–63 = format/pace features (reaction memes, deep-fried, wholesome-comic, text-post, video-clip...).
- **Music vector (128-d)**: dims 0–49 = 50 top-genre affinities (from Spotify genres + Last.fm tags, normalized); dims 50–113 = 64-d artist-embedding aggregate (co-listen collaborative signal); dims 114–127 = behavioral features (discovery appetite, nostalgia skew, mainstream index, listening-session receptivity).
- **Meme style vector**: each meme carries a 64-d vector in the same space — Phase 1 set from its human taxonomy tags, Phase 2 from CLIP image + OCR-text (Sentence-BERT) joint embedding, as v1 specified.

6.2 Incremental updates (the cheap, frequent kind)

```

on event e with weight w (laugh=1.0, save=1.2, share=1.5, meh=-0.6, skip=-0.2,
duel_reveal_resonance=2.0, co_listen=1.8, daily_drop ×1.5 multiplier): humor_vec :=
normalize( 0.995 · humor_vec + w · meme.style_vec ) -- 2% decay per ~40 events music_vec :=
normalize( 0.997 · music_vec + w · track.embedding ) humor_style := renormalize axes from dims
0–3 All inside one Edge Function transaction; vector_version++ for auditability.

```

6.3 Taste Twins score (0–100, displayed as %)

```
H = cosine(humor_vec_a, humor_vec_b) -- humor alignment
M = cosine(music_vec_a, music_vec_b) -- music alignment
S = jaccard(top_artists_a, top_artists_b) -- concrete shared-artist signal
C = calibration_confidence(min(events_a, events_b)) -- ramp 0.5–1.0 over first 50 events
P = anti_genre_penalty -- -0.25 max: b's top genres ∩ a's anti-genres
V = vibe_report_penalty -- -0.10 if a vibe-reported a lookalike cluster
raw = 0.45·H + 0.35·M + 0.20·S + P + V
TasteTwins = round( 100 · C · sigmoid( 6·(raw - 0.5) ) )
Weights are config (system_config table), versioned per vector_version — tunable without app releases.
```

- **Humor weighted above music (0.45/0.35):** v1 research found humor-style mismatch the stronger incompatibility predictor; aggressive-humor mismatch is explicitly penalized through P.
- **Candidate pipeline:** hard filters (distance, age range, orientation, mode, not-blocked, verified) → pool ~2k → score → diversify (max 2 consecutive same-top-genre cards) → freshness boost for new/active users (never a dry wasteland for newcomers) → top 200 cached.
- **Cold start:** meme calibration (15–20 swipes) + OAuth import put a user at C≈0.7 within onboarding; the Matrix shows "calibrating" copy until 10 events.
- **Fairness guardrail (from Schmooze's lesson):** gender-ratio-aware pacing — when a pool is imbalanced, throttle the over-represented side's deck velocity rather than flooding the other side; protects female retention, the metric that killed Schmooze's dating pool.

7 · API & Realtime Contract

7.1 REST / RPC endpoints (Supabase auto-REST + RPC functions)

ENDPOINT	METHOD	NOTES
/rpc/onboarding_complete	POST	Atomic: profile + calibration events + initial vectors + verification status.
/rpc/feed	POST	Cursor-paginated interleaved meme/track/drop cards; excludes seen + moderated-out.
/rpc/react	POST	{target, kind} → writes event + updates vector; idempotent by (user,target,kind).
/rpc/candidates	POST	Returns ranked deck (top 20 + cache fill) with Taste Twins + shared chips.
/rpc/decide	POST	like/pass/resonate; returns {matched, thread_id, icebreaker} on mutual.
/rpc/vibe_report, /rpc/report, /rpc/block	POST	Three distinct paths — recalibration / abuse / hard block.
/rpc/session_start end	POST	Creates session, enforces 15-min cap (free) via entitlements check.
/rpc/duel_submit, /rpc/duel_reveal	POST	Captions hidden until both submitted.
/rpc/taste_card	GET	SVG → PNG render of Fingerprint for sharing (watermarked).
/rpc/delete_account	POST	Hard delete after cooling-off; cascade purge incl. storage + push tokens.
/rpc/export_data	POST	GDPR/CCPA export: signed URL to a JSON bundle.

7.2 Realtime channels

CHANNEL	PAYLOAD	PURPOSE
<code>thread:{id}</code> (postgres_changes)	message rows	Chat delivery + read receipts
<code>thread:{id}:sig</code> (broadcast)	typing, meme-reply preview	Ephemeral signals
<code>session:{id}</code> (broadcast)	{t_ms, playing, track_id}, emoji bursts	Co-listen sync, host-authoritative
<code>user:{id}</code> (private)	new_match, duel_reveal, drop_live	In-app pings backing push

7.3 Push schedule

- **Daily Drop 9:00 AM local** — cron Edge Function per timezone shard; deep-links to the Drop card; streak-aware copy ("Don't break your 12-day streak").
- **Matches/messages** — instant, on by default. **Feed activity / marketing** — off by default.

7.4 Analytics events (PostHog)

signup_funnel(step), calibration_complete, feed_react(kind), drop_streak(n), matrix_decide(kind, taste_twins_bucket), match_created(source), first_message_latency, session_start/end(duration), duel_complete, taste_card_shared, vibe_report, paypal_view(product) — powers the phase gates in §12.

8 · Trust, Safety, Moderation & Legal

AREA	REQUIREMENT	IMPLEMENTATION (FREE)
Age	Enforced 18+	DOB collected at auth; server-side CHECK + re-check on profile edits; under-18 = hard block + appeal path.
Identity	Mandatory liveness before Matrix visibility	On-device ML Kit liveness challenge; selfie ↔ photo embedding similarity ≥ threshold in Edge Function; badge on profile; re-verify on photo change.
Text moderation	Chat + captions + prompts	Perspective API toxicity score; >0.9 auto-hold + report; contextual (flirt ≠ abuse) thresholds tuned per surface.
Image moderation	Memes, photos	Sightengine pre-publish scan (nudity/violence/CSAM-hash via PhotoDNA where available); failures → human queue.
UGC memes & copyright	DMCA Safe Harbor	Register a Copyright Agent with the U.S. Copyright Office before UGC goes live (not retroactive) ; notice-and-takedown endpoint; documented repeat-infringer policy; ranking must not specifically promote flagged content; the app never uses third-party memes in its own marketing — only licensed Meme Pack drops are external-facing.
Reporting	Store-required UGC controls	Report (abuse) / vibe-report (recalibration) / block — three separate flows; every UGC surface has report affordance; 24h SLA on abuse queue.
Safety features	Real-world dating safety	Date-share (send match profile + plans to a friend), Safety Center in settings, in-chat safety nudge on phone-number/off-platform patterns.
Data	GDPR/CCPA	Export (JSON bundle), hard delete with 14-day cooling-off, data minimization (location fuzzed to ~1km at rest), consent log.

This section summarizes the v1 legal analysis and is not legal advice — a licensed attorney should review DMCA registration, privacy policy and content policy before launch.

9 · Store Compliance & Launch Checklist

- **Apple App Store:** Sign in with Apple offered wherever Google login exists; dating-app review questionnaire; UGC moderation contact + filtering + block/report + EULA acceptance; IAP via StoreKit for all digital goods (premium, Playlist Pass, Meme Packs); account deletion in-app (mandatory); privacy nutrition labels.
- **Google Play:** UGC policy declarations, target-SDK current, Play Billing for IAP, data-safety form.
- **Legal pages:** Privacy Policy, Terms of Service, Community Guidelines (plain-language, mirrored in Safety Center), Cookie/tracking consent.
- **Ops:** support inbox, moderation console (a simple internal page over `moderation_queue`), status page, DMCA agent registered.

10 · Monetization — Free-First, Core Loop Never Gated

TIER	PRICE	UNLOCKS
Free	\$0	Full Feed + Matrix, unlimited basic messages, seeing likes (mutual reveal), 1 Resonate/day, 15-min Listening Sessions, 2 trivia plays/day, Daily Drop, Taste Card.
Premium (sub)	Monthly/ annual	Unlimited Resonates, rewind, read receipts (off by default even for subscribers), incognito, unlimited sessions & trivia, priority visibility ("SuperVibes"), advanced anti-genre filters.
Playlist Pass (IAP)	One-time	Removes the 15-min session cap without a subscription.
Meme Pack drops (IAP)	Per-pack	Licensed creator/artist meme-sticker packs — cosmetic revenue that doubles as a pre-licensed content source, sidestepping the meme copyright grey zone.
Compatibility deep-dive (IAP)	Per-match	Detailed "why you scored 87%" breakdown — premium layer over the free Taste Twins badge.

Why this split (unchanged from v1, still right): Gen Z favors micro-transactions over subscriptions, so IAPs carry impulse revenue while the subscription carries power-user perks — and neither gates a core social function. Turn Up and Boo both lost the free-tier network by paywalling likes/messages; we never do.

11 · Scaling Path — Free-Tier Limits & Upgrade Triggers

RESOURCE	FREE-TIER CEILING	TRIGGER → UPGRADE
Supabase DB (500 MB)	~30–50k users w/ lean events retention	>70% → Pro \$25/mo; archive <code>events</code> older than 90 days to storage (vectors already hold the signal).
Realtime (200 concurrent)	~2–4k DAU	>150 sustained → Pro (500 concurrent), then shard session channels.
Auth (50k MAU)	—	Over → Pro. Nothing re-architects.
Storage (1 GB) + bandwidth (2 GB)	Cloudflare CDN absorbs image traffic	Photos/voice grow first → move cold media to Cloudflare R2 (free egress) at ~5 GB.
Edge invocations (500k/mo)	~10–15k DAU	Batch vector updates (group 10 events) before paying.
Sightengine (2k ops/mo)	~60 images/day	UGC memes outgrow it → move scanning to open-source NudeNet on a \$5 VPS, keep Sightengine for reports.
pgvector ranking	100k users trivial; 1M+ needs HNSW tuning + read replica	Revenue-funded by then.

Rule: at every trigger, the fix is a plan change or a cache/archive — never a rewrite. The \$0 architecture and the \$10k/mo architecture are the same code.

12 · Build Roadmap — Phased & Gated

Each gate is a go/no-go signal check, not a date (v1's rule, kept).

PHASE	SCOPE	EXIT GATE
P0 — Foundation (weeks 1–4)	Supabase project, schema + RLS, auth (phone/Apple/Google), 18+ gate, onboarding wired to real vectors, v3 UI modularized onto real API, events pipeline + vector-updater live, basic candidates (hard filters + cosine), FCM push, PostHog.	Closed alpha: 20 friends complete onboarding; vectors visibly differentiate; crash-free sessions >99%.
P1 — Core loop (weeks 5–9)	Match Matrix + decisions + mutual-match + chat (realtime), meme pipeline (licensed seed corpus + moderation queue), Daily Drop + streak cron, Taste Twins badges + shared chips, report/block, ML Kit liveness, legal pages + DMCA agent filed.	Beta cohort (1 campus city, waitlist-gated): D1 retention ≥ 45%, ≥ 30% of matches reach a first message without Icebreaker Roulette.
P2 — Signature experiences (weeks 10–14)	Listening Sessions (sync + Our Soundtrack + 15-min cap), Meme Duel, Taste Card sharing, Icebreaker Roulette, Circles + remix, concert radar, music trivia.	Share loop works: ≥ 15% of WAU share a Taste Card/Duel card weekly; session completion ≥ 60%; store submission (Capacitor builds) passes review.
P3 — Launch & monetize (weeks 15+)	Public launch in 1–2 dense culturally-online cities (university towns first), campus/creator ambassador seeding with Meme Packs, premium + IAP live, Last.fm enrichment queue at full throughput, Apple Music connect.	1k+ DAU in launch city with healthy gender ratio; organic share-driven installs ≥ 20% of new users; free → paid ≥ 2% before expanding cities.

Appendix A · v3 Prototype → Production Mapping

V3 PROTOTYPE ELEMENT	PRODUCTION OWNER
s-feed "Today's memes", Daily Drop "Drops at 9:00 AM"	/rpc/feed + daily_drops + drop cron
s-match deck, "You and {name} resonate"	/rpc/candidates , /rpc/decide , matches + realtime notify
s-people threads, listening sessions "That's your 15 minutes"	threads/messages realtime + sessions + entitlement check
s-you meme shelf, artists, playlists, prompts, anti-genres, humor signals	profiles + fingerprints projections
Circles "From your circles"	circles/circle_posts
Onboarding "Pick a few starting tastes", "What brings you here?"	/rpc/onboarding_complete (calibration events → vectors)
localStorage state	Retired — optimistic local cache only (IndexedDB), server is source of truth

North-star metric: first-message rate within 10 minutes of match. Every v4 decision — Taste Twins badge, auto-attached icebreaker meme, Meme Duel, co-listen — exists to move that number. If a feature doesn't, it's cut.

